

Unequal- μ product floor and $n=5$ polygon floor

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Found / derived in the SharpMeow/GENChase validation campaign (2026).

Cite the author by name. Do not attribute to the GitHub handle SharpMeow.

Do not treat this note as a personal theorem stamp (no “Hendrick’s Identity / Law”).

Date: 2026-09-23 (America/New_York). Companion markdown in-repo under `research/`.

Abstract

We record exact closed forms for two classical point-vortex collapse floors: (i) the unique interior product minimum $P_\star$ on Gotoda’s $L = 0$ arc for $\Gamma = (1, 1/2, -1/3)$ with the corrected 2π Biot–Savart product; (ii) the $n = 5$ specialization $F_5 = \sqrt{31682}/80$ of the already-proved two-ring / polygon floor F_n . Both are **proved candidates**. **Priority is unconfirmed**: the expressions may already appear as classical specializations. Floor-kill body reads of Koiller et al. (1985), Aref (1982), and O’Neil (2007) return **DOES NOT KILL** for the optimized floors named here; those verdicts are limited to the texts read and do not certify novelty.

1. Statements

1.1. $\mu = 1/2$ Gotoda $L = 0$ product floor

On Gotoda’s classical $L = 0$ collapsing arc for $\Gamma = (1, 1/2, -1/3)$, with the corrected product from the 2π Biot–Savart kernel / Gotoda Prop. 2.1,

$$P(\theta) = \frac{14 \sin^2 \theta + 6\sqrt{7} \cos \theta + 21}{2(14 \cos \theta + \sqrt{7}) \sin \theta},$$

the unique interior minimum $P_\star = P(\theta_\star)$ admits the real trigonometric Cardano form

$$P_\star = \sqrt{605/324 + (7\sqrt{5201})/162 \cos(1/3 \arccos((245351\sqrt{5201})/5201^2))}.$$

Equivalently, $P_\star$ is the unique positive real root greater than 2 of the irreducible sextic

$$8748x^6 - 49005x^4 + 27794x^2 + 18723 = 0,$$

and $q = P_\star^2$ is the unique real root in $(4, 5)$ of the irreducible cubic

$$8748q^3 - 49005q^2 + 27794q + 18723 = 0.$$

Float64: $P_\star \approx 2.203855016036133$ (`tools/verify-unequal-mu-half.js`). Uses the corrected P , not the buggy Gotoda (3.3)– B specialization (≈ 1.741).

Disclaimer. Classical Gotoda unequal-circulation family; closed form for the floor, not a new interaction law.

1.1.1. Proof sketch

1. Critical cosine $\cos \theta_\star$ solves $c^3 + (8\sqrt{7}/7)c^2 + (1/14)c - 32\sqrt{7}/49 = 0$ on the collapsing arc.

2. Write $P = N(c)/(D_0(c)\sqrt{1-c^2})$ and clear radicals to a bihomogeneous relation in $(P, c, \sqrt{7})$.
3. Resultant elimination of c and $\sqrt{7}$ yields the sextic (minimal over $\mathbb{Q}$); $q = P_\star^2$ satisfies the cubic.
4. Casus irreducibilis on the cubic gives the real trig Cardano form; Float64 agrees with direct minimization.

1.2. $n = 5$ polygon-collapse floor

Specialize the classical two-ring / general polygon collapse family (Koiller et al. 1985 §11) at order $n = 5$. With the standard planar 2π Biot–Savart kernel,

$$\omega_0 t_c = \frac{127\sqrt{2} - 24\cos(5\theta)}{80\sin(5\theta)} \geq \sqrt{31682}/80, \quad 0 < \theta < \pi/5,$$

with equality uniquely at $\cos(5\theta_\star) = 12\sqrt{2}/127$. The floor is

$$F_5 = \sqrt{31682}/80 = 2.224929774172659\dots$$

under the already-proved general formula

$$F_n = \sqrt{K_n^2 - (2n-1)/(2n)}, \quad K_n = (n-1) \sinh((n+2)/2 \operatorname{arcosh}(n/(n-1))),$$

parallel to $F_2 = 3\sqrt{5}/4$, $F_3 = \sqrt{29}/3$, $F_4 = \sqrt{322}/9$. Not a new dynamical family; not a new identity row by itself.

1.2.1. Proof sketch

1. For $n = 5$, $K_5 = 127\sqrt{2}/8$.
2. Product form $\omega_0 t_c = (K_5 - 3\cos(5\theta))/(10\sin(5\theta))$ clears to the boxed quotient.
3. Elementary bound $(a - b\cos\alpha)/\sin\alpha \geq \sqrt{a^2 - b^2}$ with $a = 127\sqrt{2}$, $b = 24$ yields $\sqrt{31682}/80$.
4. Consistency: $F_5 = \sqrt{K_5^2 - 9/10}$ recovers the same radical.

2. Verification pointers

Item	Status
$P_\star$ closed form / minpoly	Algebraic derivation + Float64 (<code>tools/verify-unequal-mu-half.js</code>)
$F_5 = \sqrt{31682}/80$	Specialization of F_n ; Float64 (<code>tools/verify-new-formula-candidate.js</code>)
Priority / ledger stamp	Unconfirmed — no IDENTITIES.md promotion from this note

3. Literature (floor-kill)

Paper	Access	Verdict
Koiller et al. (1985)	Full text read	DOES NOT KILL optimized product floors

Aref, <i>Phys. Fluids</i> 25 (1982)	AIP PPV + OCR	DOES NOT KILL — center-of-symmetry / Havelock; no floor literals
O’Neil, <i>Physica D</i> 236 (2007)	ScienceDirect PPV	DOES NOT KILL — triple-ring RE/collapse; no optimized floors

Limits. Floor-kill only (absence of $P_\star$, F_5 , equality cosines, and related rate numerals in the bodies read). Not a paper-equivalence audit and not a novelty certificate. Negative search $\neq$ priority. Deeper OA greps likewise found no hard-needle hits; status remains open / priority unconfirmed.

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4. What would still kill priority

Any prior publication of the Cardano / minpoly form of $P_\star$ (or the decimal 2.203855... as that product floor), of $F_5 = \sqrt{31682}/80$ (or equivalent), of the same Gotoda- $\mu = 1/2$ minimization, or of the same K_5 clearance into $\sqrt{a^2 - b^2}$, would kill or demote priority while leaving the algebraic checks intact.

5. Suggested citation

Closed forms for the $\mu = 1/2$ Gotoda $L = 0$ product floor $P_\star$ and the $n = 5$ polygon floor $F_5 = \sqrt{31682}/80$ were derived by Chase Hendrick (2026) in the SharpMeow/GENChase validation campaign. Status: proved candidates, priority unconfirmed. Classical families only.

6. Anti-claims

- No novelty claim beyond “closed form / explicit radical written down and checked in this campaign.”
- No personal named theorem.
- No IDENTITIES.md edit from this file.
- Lit verdicts are floor-kill only.

7. References

1. H. Aref, Point vortex motions with a center of symmetry, *Phys. Fluids* **25**, 2183–2187 (1982). DOI 10.1063/1.863710.
2. J. Koiller, S. Pinto-Schuhmacher, et al., On Aref’s vortex motions with a symmetry center (1985); GENChase body read 2026-09-23.
3. K. A. O’Neil, Relative equilibrium and collapse configurations of heterogeneous vortex triple rings, *Physica D* **236**, 123–130 (2007). DOI 10.1016/j.physd.2007.07.015.
4. Y. Gotoda, $L = 0$ unequal-circulation collapse product (Prop. 2.1 / arc used in campaign drafts).
5. SharpMeow/GENChase (2026-09-23): identities/sources/unequal-mu-half-Pstar-closed-2026-09-23.md, new-formula-candidate-2026-09-23.md, Koiller / Aref / O’Neil floor-kill notes, deeper-kill-Pstar-F5-2026-09-23.md.